Introduction to FIR Filter Design

Copyright © Barry Van Veen 2014

AllSignalProcessing.com

Key Concepts

1) Advantages of FIR filter designs:

    a) FIR filters can be guaranteed to have linear phase - thus eliminating phase distortion.

    b) FIR filters are always stable.

    c) FIR filters are much less sensitive than IIR filters to numerical rounding errors in both design and implementation.

    d) The error between a desired and actual frequency response is a linear function of the unknown coefficients.

e) Optimization techniques can be used to obtain optimal designs.

2) FIR filter design criteria:

a) Minimize the mean squared error between the desired and actual frequency response.

i) Frequency sampling design.

ii) Window-based design method.

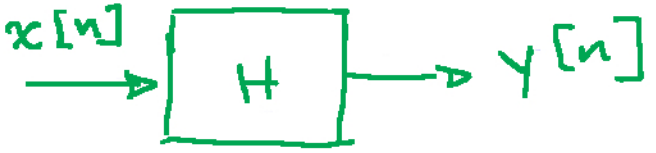
b) Minimize the maximum error between the desired and actual frequency response. Parks-McClellan design procedure.

c) Statistical filter design. Minimize the expected squared error between a desired signal and the filter output.

Video Screenshots

FIR Filters

$$y[n] = \sum_{k=0}^{M} b_k x[n-k]$$



$$H(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-j\omega k}$$

1) Can obtain exactly linear phase – no phase distortion

2) Always stable – any b_k

3) Much less sensitive to roundoff errors than IIR filters

4) Frequency response is a linear function of b_k

Optimization based designs

Desired Response: $H_d(e^{j\omega})$

Error: $E(\omega) = H_d(e^{j\omega}) - H(e^{j\omega}) = H_d(e^{j\omega}) - \sum_{k=0}^{M} b_k e^{-j\omega k}$

Error is also a linear function of b_k!

1) Minimum Mean Squared Error Design

Frequency sampling design: $\min_{b_0, b_1, \ldots, b_M} \sum_{\ell=1}^{N} |E(\omega_\ell)|^2$

2) Minimax Error Design

Parks-McClellan: $\min_{b_0, b_1, \ldots b_M} \left\{ \max_{\omega \in F} \left| W(\omega) \left(H_d(e^{j\omega}) - \sum_{k=0}^{M} b_k e^{-j\omega k} \right) \right| \right\}$

- linear phase filters -

3) Statistical filter design

$$\min_{b_0, b_1, \ldots b_M} E\left\{ \left(y_d[n] - \sum_{k=0}^{M} b_k x[n-k] \right)^2 \right\}$$

4) Window-based FIR filter design

- "ideal" infinite duration impulse response
- truncate ideal impulse response with a window
  - mainlobe ⇒ transition bandwidth
  - sidelobes ⇒ pass and stopband ripple

FIR filters are an important signal-processing tool!

For more great educational signal-processing content:

AllSignalProcessing.com

- Intuitive, easy-to-follow style
- Organization that saves you time
- Answers to your questions
- Learn at your pace
- Award winning instruction